Nate HUANG

Research Interest: Applied Optimization and Machine Learning in Engineering; Risk Analysis; Uncertainty Quantification
+1 510 501 0805 | yichai@berkeley.edu | http://linkedin.com/in/nate-huang

EDUCATION

University of California, Berkeley ***Berkeley, CA***
Master of Science in System Engineering *2027*

- Relevant Coursework: CS285: Deep RL; CS289A: Intro ML; CS227AT: Optimization Models in Eng; CE293A: Probability and Risk Analysis; CE293: Uncertainty Quantification; CE291D: Data-Driven Control Method for Eng; IEOR221: Financial Eng…

University of Leeds ***Leeds, UK***
Bachelor of Engineering in Civil Engineering with Transport *Jul 2026*

- Average Score: **71.32/100** (**First Class Honors**)

Southwest Jiaotong University (SWJTU) ***Chengdu, CN***
Bachelor of Engineering in Civil Engineering *Jul 2026*

- GPA: **3.78/4.0** | 2025 SWJTU Scholarship, 2024 SWJTU Scholarship, 2024 Outstanding Student Leader, Integrity and Excellence Award

RESEARCH EXPERIENCE

Wildfire-Induced Cellular Network Resilience via Monte Carlo–Driven Graph Neural Surrogate Modeling
291D Project Leader *Sep 2026 - Present*
Supervisor/ Instructor: Prof Maria Laura Delle Monache, UC-Berkeley

- Developing a physics and graph-informed deep learning framework to predict time-dependent cellular service outages under joint wildfire and public-safety power shutoff scenarios.
- Constructing an inferred cellular-network topology from public cell-site data and simulating direct fire damage, power loss, backup-power depletion, and network disconnection through **Monte Carlo** analysis.
- Designing a hybrid **U-Net and temporal graph neural network** surrogate to estimate component failure probabilities, probabilistic coverage maps, and **population-weighted service loss** using wildfire hazard, infrastructure, and U.S. Census data.
- Evaluating the framework against geometric coverage, Monte Carlo, CNN-only, and GNN-only baselines using **probability calibration, spatial accuracy, population-coverage error, and computational speed**.

Auditing Risk-Neutral Generative Models for Option Markets
Oct 2026 - Present

- Developing a framework for **auditing diffusion-based SPX option-surface generators and risk-neutral pricing kernels**, with a focus on model reliability and risk-premium identification.
- Built an end-to-end pipeline using **47.4 million SPX option quotes** from OptionMetrics and Heston groundtruth simulations to disentangle surface representation error, finite-sample entropy bias, and generator misspecification.
- Evaluating predictive calibration, no-arbitrage consistency and **out-of-sample pricing-kernel validity** against autoregressive and Gaussian benchmarks.

Anchor-Consistent Conditional Residual Diffusion Model for Stochastic 3D Micro-CT Completion
Undergraduate Research Assistant, *Feb 2025 - Present*
Supervisor: Prof. Ziqi Wang, Prof. Dan Tsang, Dr. etc., UC-Berkeley & HKUST

- Developed a **native-resolution conditional diffusion model** for **stochastic reconstruction** of missing slices in sparsely sampled, multi-phase GGBS CT volumes.
- Introduced a distance-conditioned residual framework to model nonlinear structural variations while preserving measured anchor slices.
- Validated reconstruction quality through morphological, topological, and cross-slice continuity metrics on held-out specimens.

Deep learning based nonlinear regular wave loads prediction for dual cylinders
Bachelor's Thesis, Superviosr: Duncan Borman from Uni of Leeds; Guoji XU from SWJTU *Dec 2025 – Jun 2026*

- Built an end-to-end physics-informed deep learning framework (**U-Net**) to predict spatiotemporal static pressure and velocity fields around dual cylinders under **5th-order nonlinear Stokes waves**.
- Conducted 3D CFD simulations in Fluent (**RANS & k-ω SST**) across multiple spacing ratios with a numerical beach; constructed an offline preprocessing pipeline to convert unstructured nodal data into standardized 256x256 multi-channel spatial tensors.
- Achieved high-fidelity physical field reconstruction (**validation MAE = 0.72%**) and speedup compared to traditional CFD solvers

Research on 3D Printed ECC Expansion Plate Preparation and Mechanical Properties
Team Leader, Supervisor: Prof. Wei He, Funded by University of Leeds *May 2024 - Sep 2024*

- Completed modeling and finite element analysis and iteratively optimized the model to obtain the optimal size of the grid.
- Led 3D printed PP - ECC bridge expansion plate design, blending advanced materials with 3D printing tech.
- Conducted comparative research on **four-point fatigue and freeze-thaw** properties of steel-reinforced, BFRP, CFRP, GFRP, and unreinforced PP-ECC materials to evaluate the materials' durability under continuous vehicle loads and extreme environmental conditions.

Model Development for Bridge Cantilever Assembly Construction Based on Digital Twins
Core Member, Student Research Training Program, Supervisor: Prof. Mingzhi Xie, SWJTU *May 2023 - Sep 2024*

- Built a 1:100 scale physical bridge model with basswood and fixed connection issues via magnets for structural stability.
- Performed **finite element analysis** on the bridge using Midas Civil and validated model accuracy via DIC experiments, resulting in an error between theoretical and actual results of **less than 10%**.

- Designed a sensor system for real-time bridge data (displacement, cable force, temp.), used SQL Server to analyze data and trigger warnings for safety and efficiency.

Fundamental Study on Compressive Creep Characteristics of Engineered Cementitious Composites (ECC)

Core Member, Supervisor: Prof. Wei He, Funded by University of Leeds *May 2023 - Sep 2023*

- Conducted an in-depth experimental investigation into the compressive creep behavior of ECC at an early age, focusing on the material's short-term performance and durability under sustained loads.
- Utilized Digital Photogrammetry for Deformation Measurement (**DPDM**) to accurately record strain and deformation.
- Established **a high-resolution image acquisition system** equipped with a telecentric lens, employed **a sub-pixel interpolation algorithm** to analyze image sequences, and obtained accurate full-field deformation data of the specimens.

PUBLICATIONS

- **Huang, Y.**, "The Application and Performance Study of New Environmentally Friendly Materials in Bridge Expansion Joint Vibration and Noise Reduction (Abstract)," in *Proceedings of the 2026 IABSE (International Association for Bridge and Structural Engineering) Symposium in Copenhagen*, 2025, Accepted.
- **Huang, Y.**, et al., "Research and Design of Highly Durable and Low-Noise ECC Bridge Expansion Joints (Abstract)," in *Proceedings of the 2025 IABSE Symposium in Tokyo*, 2025, Accepted.
- **Huang, Y.**, et al., "Comprehensive Measures and Application Research on Noise Control of ECC Bridge Expansion Joints (Abstract)," in *Proceedings of the 2025 IABSE Symposium in Tokyo*, 2025, Accepted.
- **Huang, Y.**, et al., "Research on the application of digital twin in bridge cantilever erection construction," in *Proceedings of the 2024 10th International Conference on Architectural, Civil and Hydraulic Engineering (ICACHE 2024)*, pp. 574-579, Atlantis Press, 2025. DOI: https://doi.org/10.2991/978-94-6463-658-1_58.
- Ge, R., **Huang, Y.**, et al., "Prediction Of the Future Development of New Energy Vehicles in China Based on Grey Model and Times Series," *Highlights in Science, Engineering and Technology*, vol. 93, pp.417-420, May 2024. DOI: https://doi.org/10.54097/n6zrer75.

MAIN PATENTS

- **Huang Y.**, et al., *An assembly structure for contraction joints of bridge concrete*, CN223061438U, 2025.07.04.
- **Huang Y.**, et al., *A mechanical sensor for detecting the entire life cycle of bridge expansion joints*, CN223005568U, 2025.06.20.
- **Huang Y.**, et al., *A high toughness low noise bridge concrete expansion plate*, CN222730335U, 2025.04.08.
- Shu H., **Huang Y.**, et al., *Spraying method of an environmentally friendly vibration- and noise-reducing polyurethane replacement material for bridge expansion joints*, CN119259415A, 2025.01.07.
- Liang Y., **Huang Y.**, et al., *A method for enhancing the durability of the debonding layer of concrete expansion joints*, CN119062015A, 2024.12.03

MAIN CONTEST EXPERIENCE

2024 China International College Students' Innovation Competition (Main Track)

Team Leader ***National Silver Award (Top 0.01% of 5.14 million entries)*** *Oct 2024*

- Directed a high-level **research team of 22 members** (2 academicians, 5 professors, and 15 students) to evaluate performance drawbacks in traditional steel expansion joints, focusing on noise reduction, maintenance optimization, and safety enhancements.
- Proposed a solution with **high-toughness, high-durability concrete bridge expansion plates**.
- Validated the design through laboratory testing and collaborated with a company for production, leading to successful **application** in Chengdu, CN to effectively address the shortcomings of traditional devices.

2024 American Concrete Institute International Concrete Dragon Boat Competition

Team Leader, Best Technical Proposal Award *Jun 2024*

- Designed an ECC mix ratio incorporating **glass microspheres** to enhance performance.
- Developed **a hybrid fiber reinforcement** by strategically interlacing **carbon fiber** strips into **BFRP** mesh using traditional Chinese "Fuyipanzhang" knotting techniques.
- Implemented a **water stair design** for hydrodynamic drag reduction, improving flow efficiency by 22%.
- Simulated hull pressure and wave patterns with knowledge of computational fluid dynamics.

2023 China Undergraduate Mathematical Contest in Modeling

Core Member, First Prize (Top 10%) *Sep 2023*

- Developed a scientific replenishment pricing strategy for vegetables, addressing their short shelf life and market demand fluctuations.
- Conducted data preprocessing, including one-hot encoding and time series analysis with the **SARIMA model** for sales patterns analysis.
- Employed Pearson correlation analysis and paired sample T-tests to assess relationships between product categories and sales.
- Established an optimization model using **interior point and greedy methods** to maximize profits for the replenishment strategy.

LEADERSHIP & ACTIVITIES

Operation Team Member of Engineering Structures Wechat Official Account *May 2025 – May 2026*

- Summarized and shared the **latest research** innovations published in the Engineering Structures, highlighting cutting-edge methodologies, theoretical advancements, and practical applications to foster academic exchange and broaden readership

impact.

- Shared valuable information regarding academic conferences and calls for papers, showcased leading research labs and teams.

Minister of Publicity Department, Cross-Cultural Language Exchange Association *Sep 2023 - Jun 2024*

- Organized UoL faculty and students to participate in UK traditional Countdown activities, fostering engagement and cultural exchange.

Director of Foreign Affairs and General Department, Young Volunteers Association of SWJTU-LEEDS *Sep 2023 - Jun 2024*

- Organized **106 volunteer services for 578 students** during the two years of service.